

Sarvam Sheth

US Citizen | +1 908-693-4010 | me@sarvamsheth.com | Portfolio: <https://tinyurl.com/6zkp5nm3>

Education:

University of Colorado, Boulder - B.S Aerospace Engineering | GPA: 3.7 | Dean's List | Aug 2024 - May 2028

Coursework: Statics/Dynamics, Thermodynamics, Lin-Alg, Diff-Eq, Astrophysics, Astrodynamics, Aerospace Communications

Skills:

- Designing: Solidworks | AutoCAD | 3D Printing | Electrical Wiring | ANSYS | FEA | Composites
 - Programming: MATLAB | C++ | Python | Arduino | Java | LabView
 - Other: Experimental Testing | Microsoft Office | Communication | Detail-Orientated | Teamwork
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Experience:

CU Boulder Formula SAE: Buffs Racing - *Aero Testing Lead & Aerodynamics Engineer* | Aug 2024 - Present

- Designing and iterating sidepod and undertray geometries in Onshape and SolidWorks, running CFD to increase heat transfer of radiator and downforce-to-drag ratio whilst **reducing aero package weight by 30%** under FSAE regulations.
- **Independently designed and 3D-printed** an on-car pressure sensor grid to experimentally validate CFD predictions, predict thermal models, handling sensor selection, sensor housing, NI cDAQ data acquisition, and MATLAB analysis.
- Developed controlled on-track test procedures and **technical documentation** to correlate experimental pressure distributions with simulation results and refine aero models for sidepods and other aero elements,
- Manufacturing aerodynamic components through hands-on composite fabrication, including carbon fiber and fiberglass layups with foam cores, molds, and many finishing processes in lab and workshop environments using proper PPE.
- Collaborating with chassis team, tuning suspension travel to maximize ground-effect of front wing and reduce scraping.

FRC Team 1403 Cougar Robotics - *Design Captain* | Sept 2021 - Jun 2024

- Designed mechanical systems including telescoping arms, swerve drivetrain modules, and gearboxes using Autodesk Inventor and FEA tools, iterating across time-critical cycles.
- Led development of **3D printed** lead-screw-driven telescoping arm **achieving <1 sec extension, >5 ft reach to maximize scoring** efficiency of high-reach game piece nodes whilst allowing easy maintenance after hard robot use
- Led team coordination by delegating tasks and fostering communication to enhance collaboration and productivity, **cutting production time by 50%** and **ranking 146th of 3290 globally**, highest in team history.

Astrodynamics Research - *NASA Gravity Field Anomaly Measuring* | Sept 2025 - Present

- Implemented **orbital propagators** in **MATLAB** and **GROOPS** to analyze position, velocity, and timing errors to create accurate models and predict time dilation and thus measure and map gravity anomalies on Earth
- Developing mission concepts using atomic clocks and relativistic effects to measure and predict climate change effects, **targeting reduced cost and failure modes** compared to previous GRACE missions.
- Modeling orbital motion in ECI and ECEF frames, accounting for Earth rotation and providing accurate visuals of orbits.

Princeton Airport - *Aircraft Mechanic Intern & Commercial Pilot* | May 2025 - August 2025

- Diagnosed mechanical, electrical, and avionics issues on general aviation aircraft by analyzing system behavior, failure modes, and maintenance documentation.
 - Completed a full engine replacement on a Cirrus SR20, verifying performance and logging for FAA compliance.
 - Evaluated component wear, tolerances, and operational limits, applied safety margins, reliability, and design constraints.
 - FAA Commercial Pilot with Instrument Rating; **logged 250+ flight hours**, reinforcing disciplined decision-making
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Engineering Projects:

Foam Glider Project- *ASEN 2502 Project*

- Designed and prototyped a foam glider using stepped-airfoil design to reduce skin friction whilst simplifying build and using CFD to verify improvement, setting class record of 83 meter being launched from 4th floor of building

Aerodynamic Control System - *Rocketry Club*

- Designed and built a 3D-printed four-flap drag control system to limit rocket altitude to 850ft.
- Implemented onboard control logic to adjust extension based on speed, height, using a single-motor design